TI TIDA-060029 analog front end

Matched RG/RF: 100R / 5k / 100k

File: afe.kicad_sch

DDS excitation and reconstruction

100Hz-100kHz small-signal excitation

File: source.kicad_sch

Simultaneous voltage / current acquisition

Two channels / one conversion instant

File: adc.kicad_sch

USB, RP2040 and boot / debug

USB control, firmware and host calibration

File: digital.kicad_sch

Power: 5V input, forced PWM, quiet analog rails

External 5V → quiet +/-12V

File: power.kicad_sch

Engineering prototype. One PCB. Isolated, discharged DUT only.

Temper

Sheet: /
File: temper-lcr.kicad_sch

Title: Temper LCR — TIDA-060029 based

Size: A3	Date: 2026-10-20	**Rev: C**
KiCad E.D.A. 10.0.4		Id: 6/6

U6A source buffer; U6B HI sense; U7A transimpedance; U7B LO sense.
Fit ONE MATCHED pair: JP1+JP4=100R; JP2+JP5=5k; JP3+JP6=100k.
Power OFF to change ranges. One local C104=100pF compensates all ranges.
No BUF634A: maximum source current approximately 18mA. Isolated, discharged DUT only.

TI source and range network

JP1
R40
RG 100R
JP2
100R 0.1%
R41
RG 5k
JP3
5k 0.1%
R42
RG 100k
100k 0.1%
RG100
RG5K
RG100K
U6A
EXCITE
OUT
OPA2810IDGKR
SOURCE_BUF

J3
DUT_HI
DUT_LO
DUT HI / LO

J3: connect external, isolated DUT between HI and LO.
No DUT component is fitted on this PCB.

TI transimpedance topology: Rev C compensation

C104=100pF directly across U7.1 and U7.2 in all ranges

R43
JP4
100R 0.1%
R44
RF 100R
JP5
5k 0.1%
R45
RF 5k
JP6
100k 0.1%
RF 100k
RF100
RF5K
RF100K
C104
100pF C0G 5%
U7A
GND
TIA_OUT
OPA2810IDGKR

U6B
HIGH_BUF
OPA2810IDGKR

U7B
LOW_BUF
OPA2810IDGKR

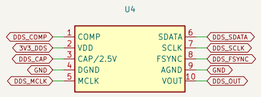
Independent HI / LO voltage followers

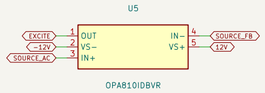
U6C
U7C
OPA2810IDGKR
12V
-12V
C42
C43
C44
C45
C50
C51
100nF 50V
1uF 50V
10nF 50V
C46
C47
C48
C49
C52
C53

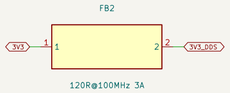
Retain TI 100nF + 1uF bypass per rail; added 10nF at each supply pin.
PCB plane windows and short feedback routes are specified in reference-conformance.md.

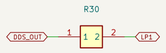
Temper
Sheet: /TI TIDA-060029 analog front end/
File: afe.kicad_sch
Title: TI TIDA-060029 analog front end
Size: A2 | Date: 2026-10-20 | Rev: C
KiCad E.D.A. 10.0.4 | Id: 2/6

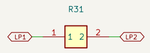
24MHz DDS clock from RP2040 120MHz / 5: sine output only.
Two 330R/1nF RC stages, AC coupling, OPA810 noninverting gain 6.
This block is added to the TI reference. Measure actual amplitude and distortion.

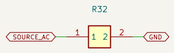


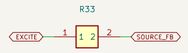


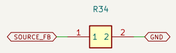














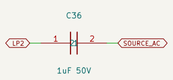




















Global net names connect functional sheets. See TI conformance matrix.

Temper

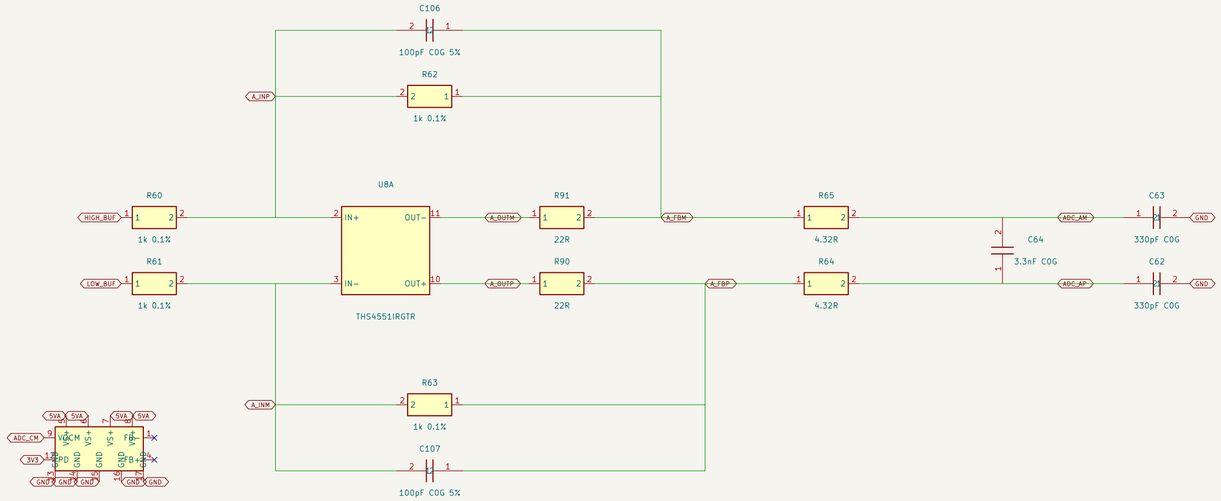
Sheet: /DDS excitation and reconstruction/
File: source.kicad_sch

Title: DDS excitation and reconstruction

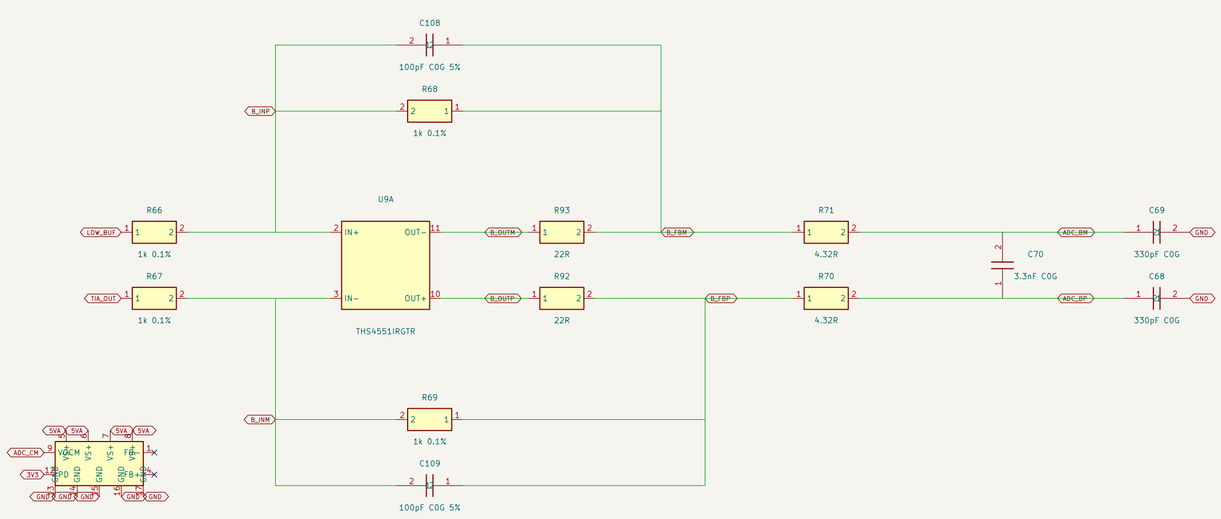
Size: A2	Date: 2026-10-20	**Rev: C**
KiCad E.D.A. 10.0.4		Id: 3/6

Channel A = HIGH_BUF – LOW_BUF; channel B = LOW_BUF – TIA_OUT.
Z = [RF / (1+j*2*pi*f*RF*CF)] * A/B. CF=100pF on every range; high-range pole=15.9kHz.
ADS9224REVM p26 topology: 22R inside feedback, 100pF || 1k, then 4.32R to ADC.
REFby2 drives VOCM. REFOUT is decoupled only. Each REFP uses 22uF.

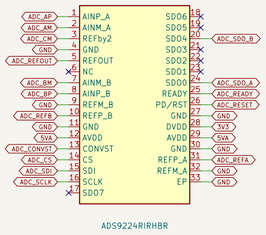
U8: feedback senses the nodes AFTER 22R, BEFORE 4.32R.



U9: feedback senses the nodes AFTER 22R, BEFORE 4.32R.

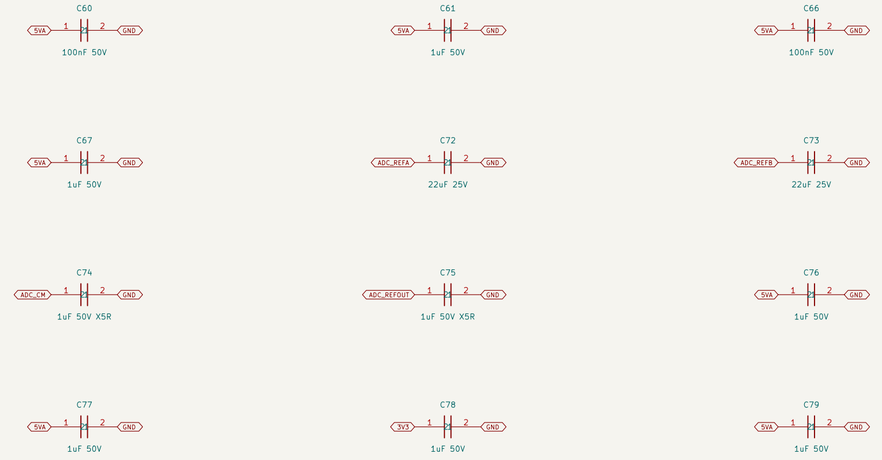


U10



ADS9224RIRHBR

Reference and supply bypass. REFby2 directly drives both VOCM pins;
C74=1uF. EVM per-driver 100R / 100nF VOCM filters are omitted.
RGT internal FB pads are NC; external feedback uses main OUT pads.

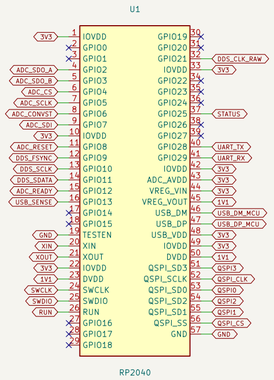


Temper
Sheet: /Simultaneous voltage / current acquisition/
File: adc.kicad_sch

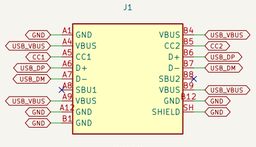
Title: Simultaneous voltage / current acquisition

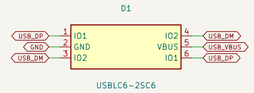
Size: A2	Date: 2026-10-20	**Rev: C**
KiCad E.D.A. 10.0.4		Id: 4/6

External 5V input powers the instrument. USB VBUS is sensed only; no backfeed.
Native USB CDC; 120MHz MCU; dual-lane ADC acquisition at up to 1MSPS.
Hold BOOTSEL while resetting for ROM USB firmware loading. SWD and UART provided.

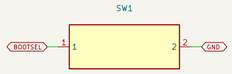


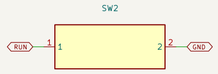


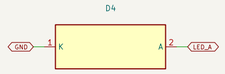


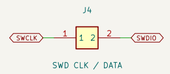


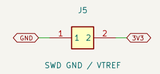


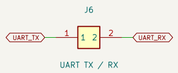


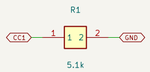


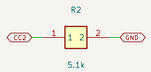


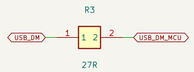


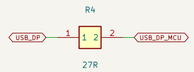


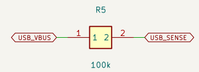


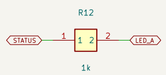


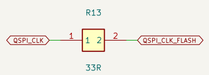










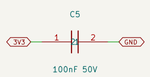




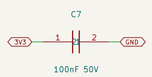


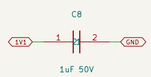










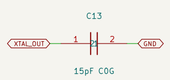




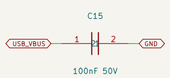












Global net names connect functional sheets. See TI conformance matrix.

Temper

Sheet: /USB, RP2040 and boot / debug/
File: digital.kicad_sch

Title: USB, RP2040 and boot / debug

Size: A2	Date: 2026-10-20	**Rev: C**
KiCad E.D.A. 10.0.4		Id: 5/6

Input: REGULATED 5V +/−5%, 1A, center positive. USB does not supply this board.
TPS65131 PSP/PSN grounded: forced PWM 1.25−1.5MHz. RAW_P −14.31V: RAW_N −−14.39V.
TPS7A49 −+11.97V: TPS7A30 −−12.00V. Fixed output values are nominal.
Do not substitute capacitor voltage/package without checking DC−bias derating.

U3



3.3V AP2112K

U11



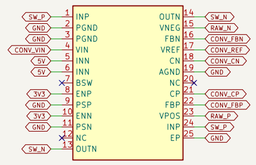
TPS7A4901DGNR

U12



TPS7A3001DGNR

U13



TPS65131RGER

Q1



AO3401A

D2



SS14

D3



SS14

L1



4.7uH 3.8A shielded

L2



4.7uH 3.8A shielded

FB1



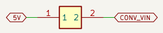
120R@100MHz 3A

J2



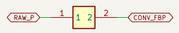
5V 1A CENTER+

R80



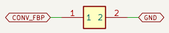
100R

R81



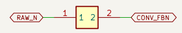
1.62M

R82



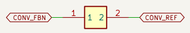
150k

R83



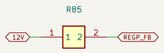
1.78M

R84



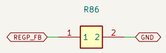
150k

R85



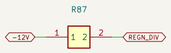
91k 0.1%

R86



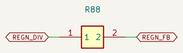
10k 0.1%

R87



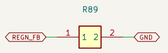
91k 0.1%

R88



1k 0.1%

R89



10k 0.1%

C16



1uF 50V

C17



1uF 50V

C18



10uF 50V

C80



10uF 50V

C81



100nF 50V

C82



10uF 50V

C83



100nF 50V

C84



220nF 50V

C85



4.7nF 50V

C86



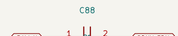
10nF 50V

C87



4.3pF C0G

C88



4.3pF C0G

C89



10uF 50V

C90



10uF 50V

C91

10uF 50V

C92



10uF 50V

C93

10nF 50V

C94

10uF 50V

C95

10uF 50V

C96

10nF 50V

C97

100nF 50V

C98

10nF 50V

C99

10uF 50V

C100

10uF 50V

C101

10nF 50V

C102

100nF 50V

C103

22uF 25V

PWR_FLAG



PWR_FLAG



PWR_FLAG



PWR_FLAG



PWR_FLAG



PWR_FLAG



PWR_FLAG



PWR_FLAG



Global net names connect functional sheets. See TI conformance matrix.

Temper

Sheet: /Power: 5V input, forced PWM, quiet analog rails/
File: power.kicad_sch

Title: Power: 5V input, forced PWM, quiet analog rails

Size: A2	Date: 2026−10−20	**Rev: C**
KiCad E.D.A. 10.0.4		Id: 6/6